Find a path in a maze

1. Statement

Given a maze in the form of a rectangular matrix, find a path from a given source to a given destination. The path can only be constructed out of cells that represents a wall and without going out of bounds; at any moment, we can only move one step in one of the four directions: Up, Right, Down, and Left, in other words, using a four neighbor topology and counterclockwise.

The main variations of this problem are: 1) using eight neighbor topology (adding diagonal movements), and 2) find the sorthest path.

2. Input

The input contains only one test case as described below.
The first line contains seven integers, R and C (1 <= R, C <= 1000) specifying the number of rows and columns, Sr, Sc, Tr, and Tc that represents the coordinates of Source cell (Sr, Sc) and Target cell (Tr, Tc). These coordinates are top left corner and 0-indexed. And finally, a number O that describes the type of output.

This is followed by the maze as a rectangular matrix M composed by R lines each one with C characters. Each character represents a cell that can either be ‘ ‘ (the space that represents a corridor) or ‘#’ (the hashtag that represents a wall).

3. Output

The meaning of the number O is:

1: “True” if exist a path or “False” other case.
2: The length as an integer.
3: The path using the four letters that represent the directions taken: U, R, D and L.

4. Sample Input

```
10 10 1 1 8 8 1
##########
#     #  #
# #   #  #
# ### #  #
#   # #  #
#   #   ##
# ###    #
```

```
# #   #  #
#     #  #
##########
```

5. Sample Output

```
True
```

Using the same maze and if the O number is 2 the output is:

```
24
```

If the O number is 3 the output is:

```
RRRRDDDDRRUUUURDDDLDDDDR
```

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